



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: Third Year B. Tech.

Course: Fuel Cell Engineering

Course Code: PEC-CH309

Date: 09-Oct-2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. All questions are compulsory.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q. N.			Marks	CO	BT
Q.1	a	What is Gibbs free energy and how is it related to spontaneity of a chemical reaction?	4	CO3	BT-I
	b	What is reaction kinetics? Explain the different types of chemical reactions?	4	CO3	BT-I
	c	Define a polarization curve and illustrate it with a suitable diagram?	4	CO3	BT-I
Q.2	a	Write about the PEM fuel cell process design, including its components, materials, properties, and process?	4	CO4	BT-II
	b	Define fuel cell efficiency and describe the different types of fuel cell efficiencies?	4	CO4	BT-I

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Basic of electrochemical energy conversion, History and development of fuel cells.
CO 2	Fuel Cell Types Components and material, properties and process
CO 3	Differentiate between various types of fuel cells and their respective advantage, limitations, and applications
CO 4	Design and analyse fuel cell components and operation conditions for enhanced performance.
CO 5	Assess fuel processing methods for hydrogen production and alternative fuel utilization in fuel cells.
CO 6	Critically evaluate the challenges, economic feasibility, and future prospects of fuel cell technology in a sustainable energy system



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: TY

Course: Material Science and
Engineering (Elective)

Course Code: PEC-CH305

Date: 09.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. All questions are compulsory
- ii. Figures to right indicate full marks.

Q.N.		Question	Marks	CO	BT
Q.1	a	Define non-ferrous metals? Explain Aluminium (Al) metals mechanical, thermal and electrical properties and industrial applications	05	CO2	BT1
Q.2	a	Explain the detail application of following non-ferrous metals I. Cobalt (Co) II. Nickel (Ni)	05	CO2	BT2
Q.3	a	What is pig iron? Explain type, properties and application of pig iron.	05	CO1	BT1
Q.4	a	What is steel? Explain classification of steel.	05	CO1	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Identify and differentiate between various ferrous metals based on their properties and uses.
CO 2	Explain the significance and applications of non-ferrous metals in engineering and industry.
CO 3	Interpret phase diagrams and understand the strengthening techniques of metal alloys.
CO 4	Describe the classification, composition, and industrial significance of glass.
CO 5	Analyze polymer properties, polymerization methods, and their applications in different fields.
CO6	Evaluate the properties and applications of composite and nanomaterials in modern technology



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
ISE-II- Examination (Semester-I) 2025-26

Department : Chemical Engineering
Chemical Reaction Engineering -I

Class : Third Year
Sub. Code: PCC-CH305
Max. Marks :20

Date: 11/10/2025

Time: 11:00 -11:45

- Note:** i. Attempt any four questions
 ii. Assume suitable data if necessary
 iii. Use of non-programmable calculator is permitted.

Q.N	Questions	Marks	CO	BT
Q 1	Give the broad classification of reactor types with sketch and also write material balance for an element of volume	05	3	1&2
Q 2	Consider a feed $C_{AO} = 100$, $C_{BO} = 200$, $C_{IO} = 100$ to a steady-flow reactor. The isothermal gas-phase reaction is $A+3B \rightarrow 6R$. If $C_A = 40$ at the reactor exit, what is C_B , X_A , and X_B there? $C_{AO} = 100$ $C_{BO} = 200$ $C_{IO} = 100$  $C_A = 40$ $C_B = ?$ $X_A = ?$ $X_B = ?$	05	3	5
Q 3	Derive performance equation for steady-state plug flow reactor along with graphical representation of the performance equations for plug flow reactors.	05	2	3&4
Q 4	An aqueous feed of A and B (400 liter/min, 100 mmol A/liter, 200 mmol B/liter) is to be converted to product in a plug flow reactor. The kinetics of the reaction is represented by $\text{mol } A+B \rightarrow R$, $-r_A = 200 C_A C_B \text{ liter.min}$. Find the volume of reactor needed for 99.9% conversion of A to product.	05	3	5
Q 5	Explain size comparison incase of Mixed Versus Plug Flow Reactors.	05	4	3
Q 6	A liquid reactant stream (1 mol/liter) passes through two mixed flow reactors in a series. The concentration of A in the exits of the first reactor is 0.5 mol/liter. Find the concentration in the exits stream of the second reactor. The reaction is second order with respect to A and $V_2/V_1=2$	05	3	5

Course Outcomes (CO)

CO-1	Understand fundamental concepts of reaction engineering, type of reactions, and reaction kinetics.
CO-2	Enhance their knowledge of reactors, the working of reactors, and different types of arrangements of reactors.
CO-3	Improve the student's ability to decide the type of reactor and their arrangement or achieving the desired conversion while keeping the economy in point of view.
CO-4	Analyze series and parallel reactions, compute yield/selectivity, and design reactors accordingly.
CO-5	Solve energy balance equations and evaluate the impact of temperature variation on reactor performance
CO-6	Solve numerical and conceptual problems in reaction engineering, preparing for industrial applications and advanced studies.

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Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: TY

Course: Numerical Methods in
Chemical Engineering

Course Code: PEC-CH304

Date: 09.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. All questions are compulsory
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.

Q.N.		Question	Marks	CO	BT
Q.1	a	Use Newton Rapson Method to find the root of $2x^2 + 5 = e^x$ accurate to five decimal places, considered starting point $x_0 = 3.5$	05	CO1	BT1
Q.2	a	Find out the roots of the equation $x^3 - x - 4 = 0$ using Regula false method, $x_0 = 1$, $x_1 = 2$.	05	CO2	BT2
Q.3	a	Examine a real root of the equation $x^2 - 3x + 1$ by using iteration method	05	CO1	BT3
Q.4	a	List a real root of the equation $x^3 - 5x + 3$ by using Bisection method up to three decimal digits, 10th iteration	05	CO2	BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply numerical methods to solve problems involving material and energy balances, fluid flow operations, heat and mass transfer.
CO 2	Determine roots of algebraic equations, solution of simultaneous equations and ordinary differential equations.
CO 3	Solve problems using regression analysis, interpolation, extrapolation and numerical differentiation and numerical integration
CO 4	Able to write Computer programming in modular form & applicability of subroutines libraries to design modules



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: TY B. Tech.	Subject: Mass Transfer II	Subject Code: PCC-CH303
Date: 10.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
Q.1	Define following terms 1. Moisture content (wet basis) 2. Critical moisture content 3. Bound moisture 4. Humid heat 5. Approach in the cooling tower 6. Cooling range of cooling tower	6	CO3&4	BT1&2
Q.2	Develop the equation for the drying time required for a process having a constant rate drying period followed by a falling rate period, in which the drying rate decreases linearly with the moisture content. The final expression should include the equilibrium moisture content (x^*).	7	CO3	BT3
Q.3	It is planned to cool water from 43.3 °C to 29.4 °C in a packed counter current water-cooling tower using entering air 29.4 °C with a wet bulb temperature of 23.9 °C. The water flow is 9764.9 kg/h.m ² and the air flow is 6835.4 kg/h.m ² . The overall mass transfer coefficient is $K_Y'a = 2500 \text{ kg/h.m}^3 (\Delta Y')$. Estimate minimum air rate that can be used. Given: Enthalpy balance: $L \times C_{WL}(T_{L2}-T_{L1}) = G_s \times (H_2'-H_1')$ $H' = (1.005 + 1.88 Y') (T_G - T_0) + 2500 Y'$ $\ln P_A^V (\text{bar}) = 11.96481 - (3984.923 / (T - 39.724))$; T is in K $Y' = (P_A^V \times 18.02) / ((P - P_A^V) \times 28.97)$	7	CO4	BT5

== ALL THE BEST ==

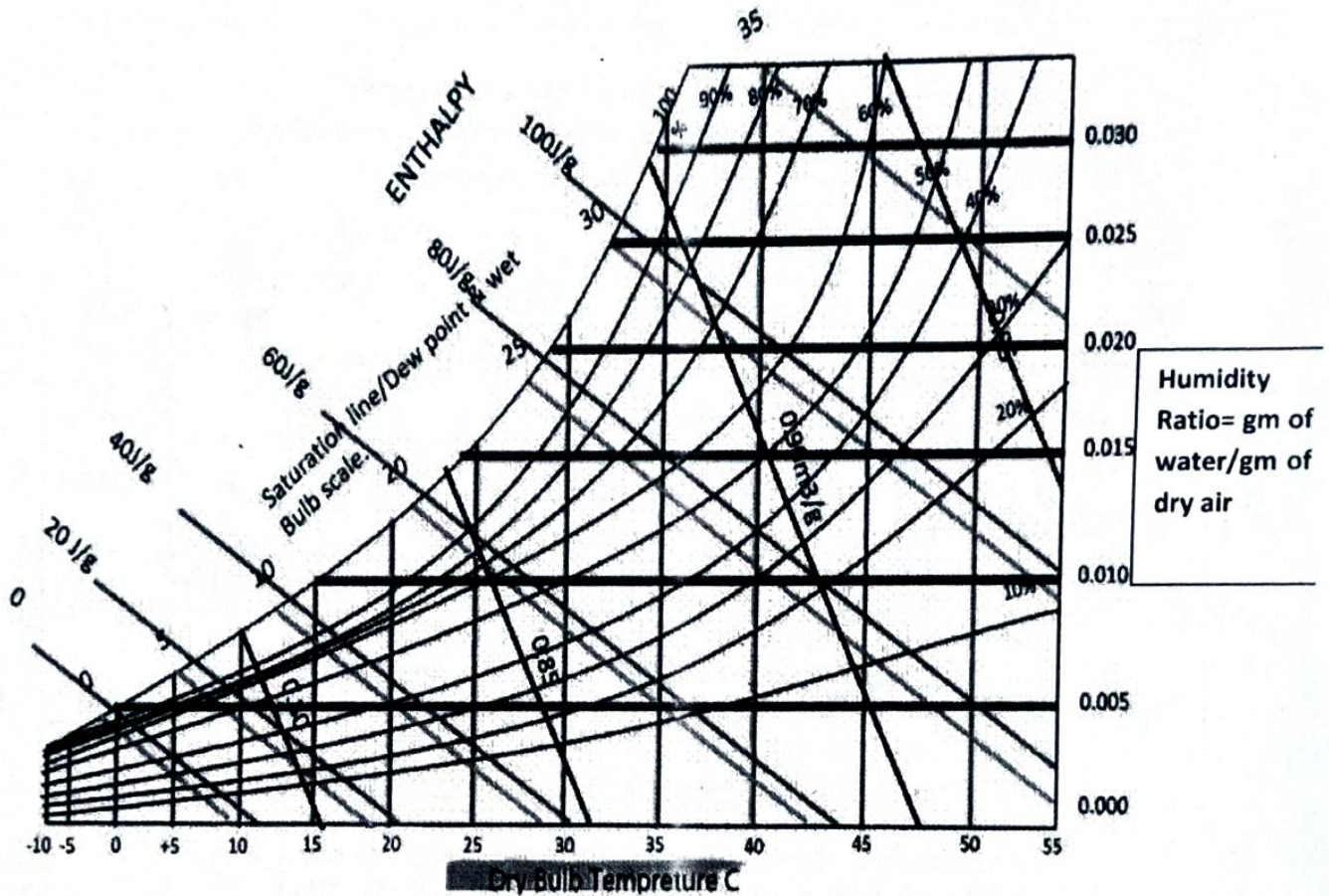
Bloom's Taxonomy (BT)

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Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Analyse gas-liquid equilibrium of absorption and stripping towers
CO 2	Design absorption and stripping towers using mass transfer correlations and stage-wise calculations.

CO 3	Evaluate drying kinetics and apply drying rate data to estimate drying time and select appropriate drying equipment.
CO 4	Interpret psychrometric charts and perform cooling tower calculations for humidification and dehumidification applications.
CO 5	Understand and apply crystallization principles to design batch and continuous crystallization processes.
CO 6	Explain membrane separation processes, including pressure-driven and concentration-driven techniques, and their industrial significance.





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Department of Chemical Engineering
In-Semester Examination-II
Academic Year: 2025-26 (semester I)

Class: Third Year	Course: Plant Utility	Course Code: PEC-CH303
Date: 09/10/2025	Time: 11.00 to 11:45	Max. Marks: 20

Note:

- Solve any Two questions.
- Figures to right indicate full marks.

Q.N.			Marks	CO	BT
Q.1	a	Enlist the boiler mountings and accessories and explain the role of each.	08	CO1	BT1
	b	What do you understand by the term Steam Trap?	02	CO1	BT2
Q.2	a	Explain the steam distribution system in any process plant with neat labelled diagram.	08	CO2	BT2
	b	Write the requirements of a good boiler	02	CO2	BT1
Q.3	a	Mention the role and importance of Compressor in industry.	05	CO2	BT2
	b	Write notes on: Economiser.	05	CO1	BT1

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Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand Industrial Water Treatment – Explain water properties, hardness, boiler feed issues, and water softening methods.
CO 2	Explain Steam & Non-Steam Heating Systems – Describe steam properties, boilers, mountings, accessories, and Indian Boiler Act.
CO 3	Analyze Refrigeration Systems – Understand refrigeration concepts, cycles, refrigerants, and performance evaluation.
CO 4	Apply Psychrometry Principles – Interpret psychrometric properties, processes, and cooling tower operations.
CO 5	Evaluate Insulation Techniques – Explain insulation materials, their importance, and applications in process industries.
CO 6	Understand Plant Maintenance Strategies – Learn types of maintenance, responsibilities, and online equipment maintenance.